

College AI Program: Advanced AI Applications & Architecture

Goal: Transition from basic conversational web windows to production-ready API engineering layouts, enterprise data structures, and multi-agent systems.

Milestone 01: Foundational Ecosystems & API Hub Architectures (Classes 1-9)

Focus: Moving past consumer UIs to programmatically control LLMs, configure token budgets, and manage context layers.

Session	Engineering Core	Syllabus Blueprint & Deliverables
Class 1	The Shift to Generative Architectures	The transition from classical ML to LLMs. Architecture of transformers, embeddings, and context limits.
Class 2	The AI Cloud Landscape & Model Selection	Comparing frontier systems (Anthropic, OpenAI, Google). Cost vs. performance vs. latency optimizations.
Class 3	System-Level Prompt Design	Mastering structured text protocols (XML, markdown delimiters) and role-setting for deterministic parsing.
Class 4	Introduction to programmatic APIs	Setting up local development runtimes (Python/Node) and establishing authentications with cloud providers.
Class 5	Anatomy of an Inference Request	Deconstructing backend request headers, payloads, temperature controls, and raw JSON responses.
Class 6	Token Lifecycle & Cost Management	Tracking token utilization, implementing input/output optimization strategies, and managing cost boundaries.
Class 7	Context Window Engineering & Caching	Deep dive into context window economics, text chunking, and implementing prompt-caching layers.
Class 8	Resilient API Pipelines	Building enterprise-grade error-handling frameworks, rate-limit mitigations, and automated retry logic.
Class 9	Milestone 1 Lab: The Engine Setup	Building a custom local CLI tool that pipes dynamic text datasets directly into frontier models via APIs.

Milestone 02: Knowledge Enrichment, Vector DBs & RAG Arrays (Classes 10-18)

Focus: Connecting enterprise external data structures safely to context tracking systems using semantic search.

Session	Engineering Core	Syllabus Blueprint & Deliverables
Class 10	The Grounding Problem & Hallucinations	Why LLMs lie. Understanding the limits of static knowledge training vs. dynamically injected context.
Class 11	Vector Embeddings Deep Dive	Transforming text blocks into high-dimensional geometric vectors. Semantic similarity vs. keyword search.
Class 12	Data Ingestion & Text Splitting	Strategies for chunking messy data (PDFs, markdown, codebases) without losing semantic intent.
Class 13	Introduction to Vector Databases	Setting up cloud and local Vector DBs (Pinecone, Chroma, or pgvector) for lightning-fast lookups.
Class 14	The Retrieval-Augmented Generation (RAG) Flow	Architecting the end-to-end RAG pipeline: Query -> Embed -> Retrieve -> Augment -> Generate.
Class 15	Advanced Retrieval Tuning	Hybrid search (Keyword + Vector), re-ranking strategies, and context compression techniques.
Class 16	Parsing Structured Tabular Datasets	Strategies for feeding complex spreadsheets, CSV arrays, and SQL records safely into the LLM context layer.
Class 17	Evaluation Metrics for RAG Systems	Measuring system factual accuracy, hallucination rates, retrieval relevance, and response quality.
Class 18	Milestone 2 Lab: Knowledge Base Build	Building an intelligent corporate knowledge assistant that reads localized internal documentation accurately.

Milestone 03: Multi-Agent Workflow Labs (Classes 19-27)

Focus: Developing decentralized code script chains where multiple processing nodes evaluate data records to isolate system flaws.

Session	Engineering Core	Syllabus Blueprint & Deliverables
Class 19	From Pipelines to Autonomous Agents	Understanding agent state machines, tool usage execution loops, and the classic ReAct framework.
Class 20	Function Calling & Schema Injection	Teaching LLMs to interact with external code functions by outputting structural tool calls.
Class 21	State Management & Cognitive Loops	Designing stateful applications where an agent logs its plan, execution history, and errors.
Class 22	Infinite Loop & Budget Guardrails	Implementing system timeouts, hard step limits, and programmatic token termination protocols.
Class 23	Introduction to Multi-Agent Frameworks	Architecting systems where specialized agents interact (e.g., Researcher -> Writer -> Code Auditor).
Class 24	The Model Context Protocol (MCP)	Bridging cloud brains with local machines. Setting up MCP clients and configuring custom tool servers.
Class 25	Human-in-the-Loop Orchestration	Inserting interactive terminal checkpoints for human confirmation before critical system tasks execute.
Class 26	Debugging Non-Deterministic Code	Strategies for tracing, logging, and troubleshooting complex multi-agent execution branches.
Class 27	Milestone 3 Lab: Collaborative Agent Squad	Developing a two-agent system where Agent A builds software modules and Agent B automatically audits them for bugs.

Milestone 04: Enterprise System Launch & Capstone (Classes 28-35)

Focus: Packaging production-grade server layouts and shipping fully operational systems into metric reporting centers.

Session	Engineering Core	Syllabus Blueprint & Deliverables
Class 28	Production-Grade App Architecture	Structuring a backend server framework (FastAPI/Express) to serve LLM and agent application requests.
Class 29	Streaming Responses & Real-Time UX	Implementing server-sent events (SSE) to stream text generation tokens continuously to client windows.
Class 30	LLM Security, Guardrails & Jailbreaks	Securing systems against prompt injection attacks, sensitive data exfiltration, and content violations.
Class 31	Monitoring & Observability Pipelines	Setting up trace tracking dashboards (LangSmith, Phoenix, or Helicone) to audit every backend LLM call.
Class 32	Capstone Kickoff & System Architecture	Mapping out individual enterprise-grade architectures incorporating APIs, RAG databases, and tools.
Class 33	Capstone Workshop: Implementation Sprint	Dedicated 1-on-1 engineering lab to code and stitch multi-layered pipeline layers together.
Class 34	System Deployment & Containerization	Packaging the final pipeline application into reproducible environments (Docker) for cloud-hosted hosting.
Class 35	Enterprise Pitch & Review Gala	Live deployment demonstration, architecture walk-throughs, and structural code reviews of the complete system.